Aftertreatment

During the engine cleaning process, ensure that water or cleaning fluids cannot enter the aftertreatment system. If cleaning fluids enters the aftertreatment system, damage could occur.

i01953476

Engine Air Filter Primary Element - Clean/Replace

SMCS Code: 1054-070-PY; 1054-510-PY

NOTICE
Service the air cleaner only with the engine stopped. Engine damage could result.

Service the air cleaner filter element when the yellow piston on the engine air filter service indicator enters the red zone or the indicator reads 63.5 cm (25 inch) of water. Refer to Operation and Maintenance Manual, "Engine Air Filter Service Indicator - Inspect".

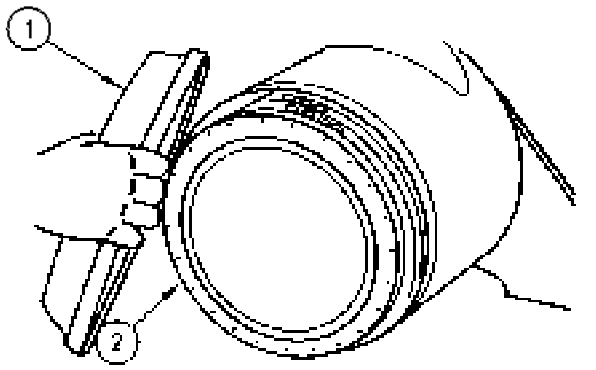


Illustration 105 g00102316

1. Remove cover (1) for the air filter housing.

2. Remove primary filter element (2) from the air filter housing.

3. Clean the inside of the air filter housing.

4. If the machine is equipped with a vacuator valve, clean the vacuator valve on the cover for the air filter housing.

5. Install a clean primary air filter element. Install the cover for the air filter housing.

Note: Refer to "Cleaning Primary Air Filter Elements".

6. Reset the engine air filter service indicator.

7. Close the access door.

If the yellow piston in the indicator moves into the red zone after starting the engine or the exhaust smoke is still black after installation of a clean primary filter element, install a new primary filter element. If the piston remains in the red zone replace the secondary element.

Cleaning Primary Air Filter Elements

NOTICE
Caterpillar recommends certified air filter cleaning services available at participating Caterpillar dealers. The Caterpillar cleaning process uses proven procedures to assure consistent quality and sufficient filter life.

Observe the following guidelines if you attempt to clean the filter element:

Do not tap or strike the filter element in order to remove dust.

Do not wash the filter element.

Use low pressure compressed air in order to remove the dust from the filter element. Air pressure must not exceed 207 kPa (30 psi). Direct the air flow up the pleats and down the pleats from the inside of the filter element. Take extreme care in order to avoid damage to the pleats.

Do not use air filters with damaged pleats, gaskets, or seals. Dirt entering the engine will cause damage to engine components.

The primary air filter element can be used up to six times if the element is properly cleaned and if the element is properly inspected. When the primary air filter element is cleaned, check for rips or tears in the filter material. The primary air filter element should be replaced at least one time per year. This replacement should be performed regardless of the number of cleanings.

NOTICE
Do not clean the air filter elements by bumping or tapping. This could damage the seals. Do not use elements with damaged pleats, gaskets, or seals. Damaged elements will allow dirt to pass through. Engine damage could result.

Visually inspect the primary air filter elements before cleaning. Inspect the air filter elements for damage to the seal, the gaskets, and the outer cover. Discard any damaged air filter elements.

There are two common methods that are used to clean primary air filter elements:

- Pressurized air
- Vacuum cleaning

Pressurized Air

Pressurized air can be used to clean primary air filter elements that have not been cleaned more than two times. Pressurized air will not remove deposits of carbon and oil. Use filtered, dry air with a maximum pressure of 207 kPa (30 psi).

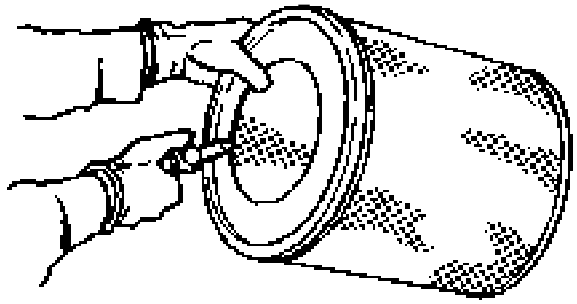


Illustration 106 g00281692

Note: When the primary air filter elements are cleaned, always begin with the clean side (inside) in order to force dirt particles toward the dirty side (outside).

Aim the hose so that the air flows inside the element along the length of the filter in order to help prevent damage to the paper pleats. Do not aim the stream of air directly at the primary air filter element. Dirt could be forced further into the pleats.

Vacuum Cleaning

Vacuum cleaning is another method for cleaning primary air filter elements which require daily cleaning because of a dry, dusty environment. Cleaning with pressurized air is recommended prior to vacuum cleaning. Vacuum cleaning will not remove deposits of carbon and oil.

Inspecting the Primary Air Filter Elements



Illustration 107 g00281693

Inspect the clean, dry primary air filter element. Use a 60 watt blue light in a dark room or in a similar facility. Place the blue light in the primary air filter element. Rotate the primary air filter element. Inspect the primary air filter element for tears and/or holes. Inspect the primary air filter element for light that may show through the filter material. If it is necessary in order to confirm the result, compare the primary air filter element to a new primary air filter element that has the same part number.

Do not use a primary air filter element that has any tears and/or holes in the filter material. Do not use a primary air filter element with damaged pleats, gaskets or seals. Discard damaged primary air filter elements.

Storing Primary Air Filter Elements

If a primary air filter element that passes inspection will not be used, the primary air filter element can be stored for future use.

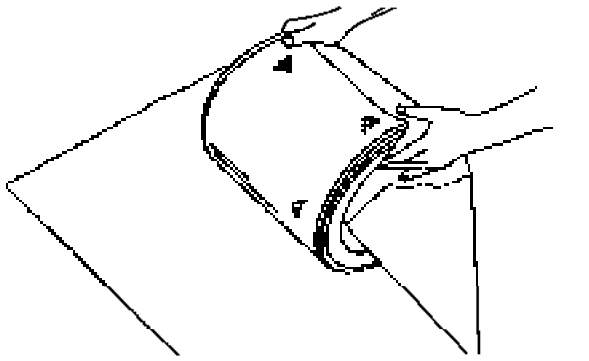


Illustration 108 g00281694

Do not use paint, a waterproof cover, or plastic as a protective covering for storage. An airflow restriction may result. To protect against dirt and damage, wrap the primary air filter elements in Volatile Corrosion Inhibited (VCI) paper.

Place the primary air filter element into a box for storage. For identification, mark the outside of the box and mark the primary air filter element. Include the following information:

- Date of cleaning
- Number of cleanings

Store the box in a dry location.

i01822684

Engine Air Filter Secondary Element - Replace

SMCS Code: 1054-510-SE

NOTICE
Always replace the secondary filter element. Never attempt to reuse the secondary filter element by cleaning the element.

When the primary filter element is replaced, the secondary filter element should be replaced.

The secondary filter element should also be replaced if the exhaust smoke is still black.

1. Open the access door on the front left side of the machine.
2. See Operation and Maintenance Manual, “Engine Air Filter Primary Element - Clean/Replace”. Remove the air cleaner cover from the air cleaner housing. Remove the primary filter element from the air cleaner housing.

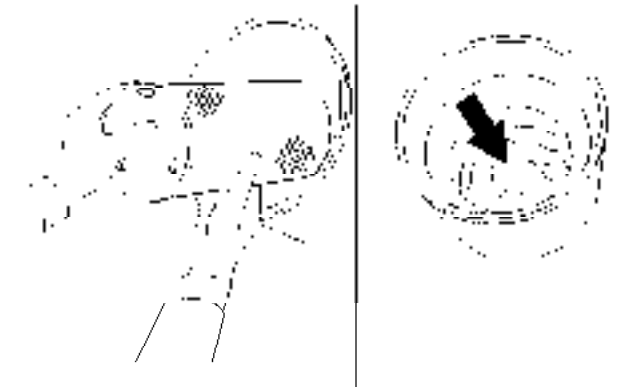


Illustration 109 g00101451

3. Remove the secondary filter element.
4. Cover the air inlet opening. Clean the inside of the air cleaner housing.
5. Remove the cover from the air inlet opening.
6. Install the new secondary filter element.
7. Install the primary filter element.
8. Install the air cleaner cover and close the latches securely.
9. Close the access door.

i02657607

Engine Crankcase Breather - Replace

SMCS Code: 1317-510

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

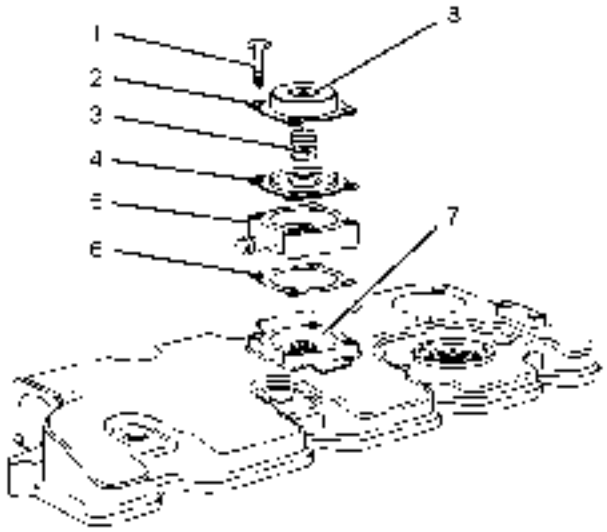


Illustration 110 g01335247

Typical example

(1) Screws for the breather cover
(2) Breather cover
(3) Spring
(4) Diaphragm and plate
(5) Spacer for turbocharged engines only
(6) Joint for turbocharged engines only
(7) Cavity
(8) Vent hole

1. Loosen the screws (1) and remove the breather cover (2) from the valve mechanism cover.
2. Remove the spring (3). Remove the diaphragm and plate (4).
3. For turbocharged engines, remove the spacer (5) and the joint (6).
4. Clean the vent hole (8) and the cavity (7) in the valve mechanism cover.

NOTICE
Make sure that the components of the breather assembly are installed correctly. Engine damage may occur if the breather assembly is not working correctly.

5. For turbocharged engines, install a new joint (6) and the spacer (5).
6. Install a new diaphragm and plate (4) for the breather assembly into the cavity (7) of the valve mechanism cover or the spacer (5) for turbocharged engines.
7. Install a new spring (3).
8. Install the breather cover (2) and the four screws (1). Tighten the screws.

i05400970

Engine Mounts - Inspect

SMCS Code: 1152-040

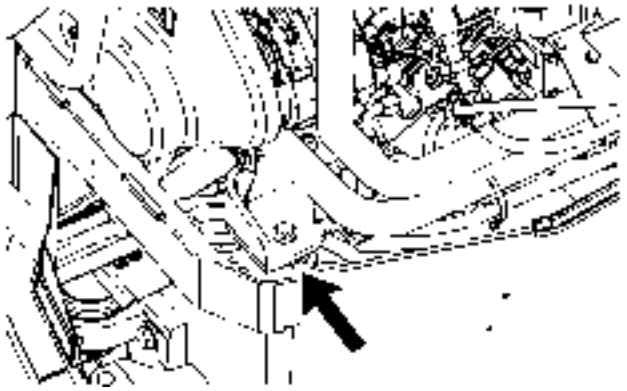


Illustration 111 g03412457

Open the engine compartment.

There are four engine mounts. There are two engine mounts on each side of the machine.

Engine vibration can be caused by improper mounting of the engine. Engine vibration can be caused by loose engine mounts or deteriorated engine mounts.

Inspect the engine mounts for deterioration.

Replace any engine mount that is deteriorated.

Inspect the engine mounts for correct bolt torque.

Tighten the mounts if the mounts are loose.

Close the engine compartment.

i08468440

Engine Oil Level - Check

SMCS Code: 1348-535-FLV

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact the skin.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Do not overfill the crankcase. Engine damage can result.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

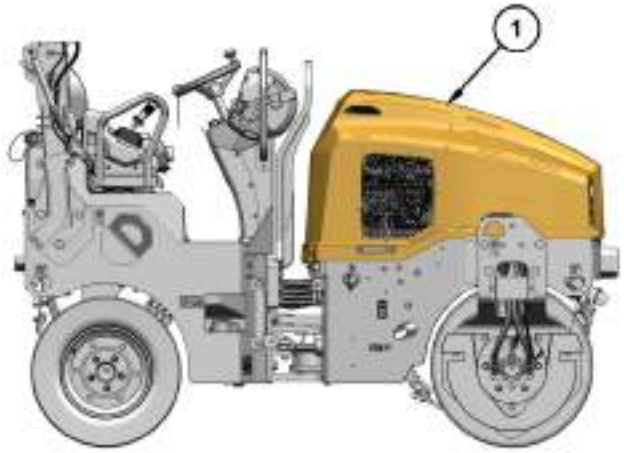


Illustration 112 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

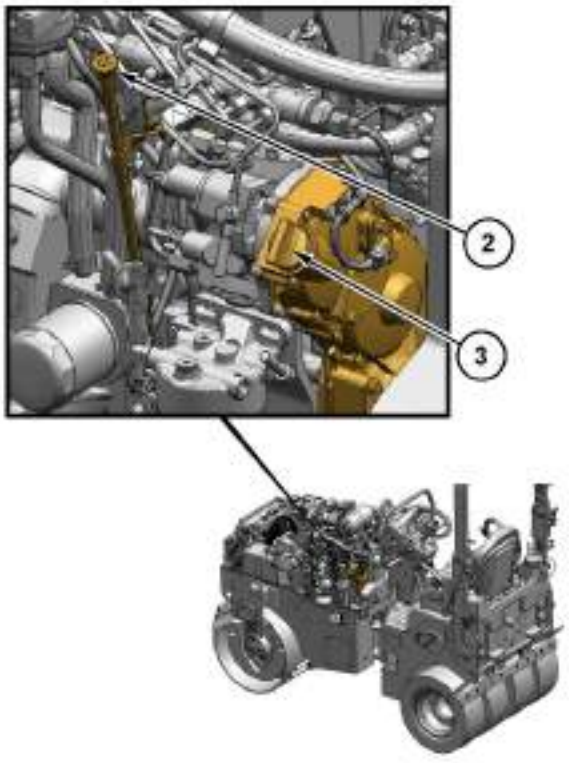


Illustration 113 g06675638

(2) Oil level gauge
(3) Oil filler cap

2. Remove oil level gauge (2) and wipe oil level gauge (2) with a clean cloth.

3. Insert oil level gauge (2) and remove oil level gauge (2) and check the oil level.

4. Insert oil level gauge (2).

NOTICE
Do not overfill the crankcase. The oil level must not reach the top of the **FULL** range mark or above the **FULL** range mark.

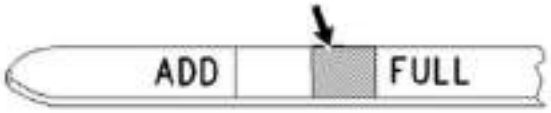


Illustration 114 g06675813

5. Maintain the oil level in oil level gauge (2) between the “FULL” and “ADD” mark . Add oil if the oil level is too low.

Note: Operating the engine with oil level above the full mark could cause the crankshaft to dip into the oil. If the crankshaft dips into the oil, the result can be excessively high operating temperatures. The high operating temperatures could result in reduced lubricating characteristics of the oil. Reduced lubricating characteristics can cause bearing damage and loss of engine power.

6. To fill engine oil. Clean oil filler cap (3) and the area surrounding oil filler cap (3).

7. Remove oil filler cap (3).

8. Add the suitable engine oil. Refer to “Lubricant Viscosities“ and “Capacities (Refill)“ for more information.

9. Clean oil filler cap (3) and install oil filler cap (3).

10. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i01175145

Engine Oil Sample - Obtain

SMCS Code: 1000-008

⚠ WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact the skin.

Obtain the Sample and the Analysis

In addition to a good preventive maintenance program, Caterpillar recommends using S·O·S oil analysis at regular scheduled intervals in order to monitor the condition of the engine and the maintenance requirements of the engine.

Each oil sample should be taken when the oil is warm and when the oil is well mixed. The sample should be taken at this time in order to ensure that the sample is representative of the oil in the crankcase.

Obtain the S·O·S Sample

Use the following method in order to obtain an S·O·S sample:

- Use a 1U-5718 Vacuum Pump or use an equivalent pump that is inserted into the sump.

To avoid contamination of the oil samples, the tools and the supplies that are used for obtaining oil samples must be clean.

Consult your Caterpillar dealer for complete information and assistance in establishing an S·O·S program for your engine.

If you fill the engine too fast with oil, the oil may saturate the engine breather. If the breather is saturated with oil, oil will blow out of the breather hose until the breather is free of oil. Add the engine oil at a rate of 2 L/min (0.5283 US gpm). This will help prevent saturating the breather with oil.

i08468502

Engine Oil and Filter - Change

SMCS Code: 1318-510

WARNING

Accidental machine starting can cause injury or death to personnel working on the machine.

To avoid accidental machine starting, turn the battery disconnect switch to the OFF position and remove the key. If the machine is not equipped with a battery disconnect switch, disconnect the battery cables from the battery and tape the battery clamps.

Place a do not operate tag at the battery disconnect switch location to inform personnel that the machine is being worked on.

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact the skin.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Do not overfill the crankcase. Engine damage can result.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

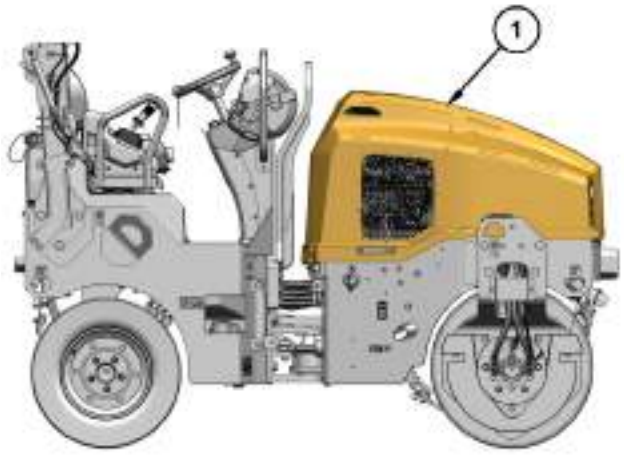


Illustration 115 g06674584

(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

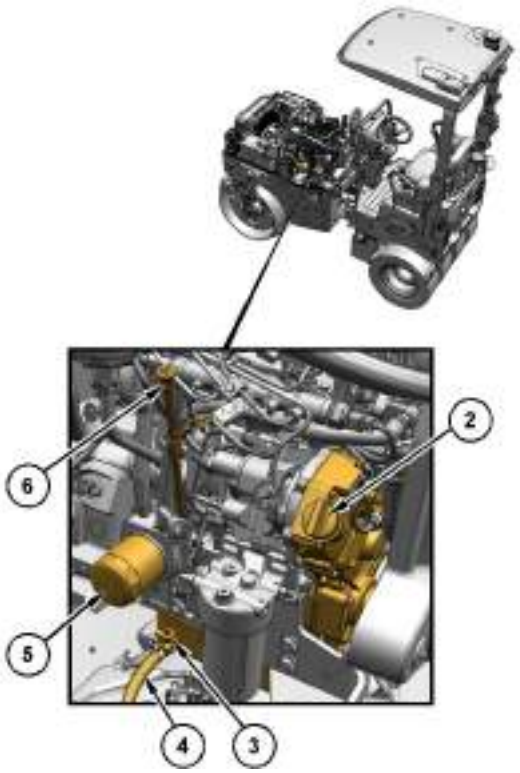


Illustration 116 g06675709

(2) Oil filler cap
(3) Drain valve
(4) Drain hose
(5) Oil filter
(6) Oil level gauge

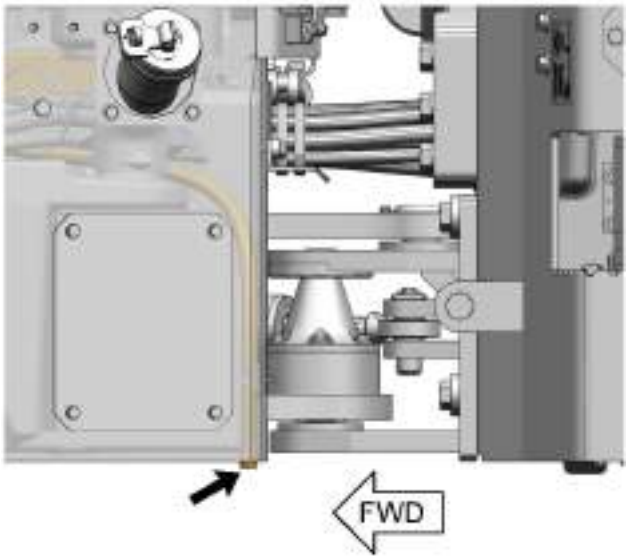


Illustration 117 g06675719

Left side view of machine showing location of drain hose outlet

2. Place a container below the drain hose (4) outlet.

3. Open drain valve (3) by turning the lever counter clockwise until the lever aligns with the outlet of drain valve (3).

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

4. Allow the oil to drain in to the container.

Note: Dispose the oil according to local regulations.

5. Close drain valve (3).

6. Remove oil filter (5).

Note: Discard the used oil filter (5) according to local regulations.

7. Clean the filter housing base. All the old filter seal must be removed from the filter housing base.

8. Apply a thin coat of engine oil to the seal of the new oil filter (5).

9. Install the new oil filter (5) by hand. When the gasket contacts the filter base, tighten oil filter (5) for an additional 3/4 turn. This will tighten oil filter (5) sufficiently.

10. Clean oil filler cap (2) and the area surrounding oil filler cap (2).

11. Remove oil filler cap (2). Fill the crankcase with new oil. Refer to “Lubricant Viscosities“ and “Capacities (Refill)“ for more information.

12. Clean oil filler cap (2) and install oil filler cap (2).

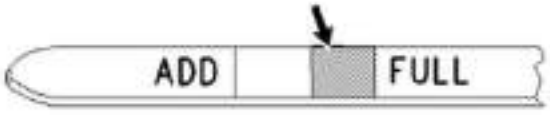


Illustration 118 g06675813

NOTICE
Do not overfill the crankcase. The oil level must not reach the top of the **FULL** range mark or above the **FULL** range mark.

13. Before starting the engine, check the oil level on engine oil level gauge (6). The oil level must be within the “FULL” range on oil level gauge (6).

14. Start the engine and run the engine for 2 minutes. Refer to “Engine Starting“ for more information.

15. Inspect the engine for any leaks. Stop the engine. Refer to “Stopping the Engine“ for more information.

16. Wait for ten minutes to allow the oil to drain back into the crankcase. Check the oil level on oil level gauge (6). Maintain the oil level within the “FULL” range on oil level gauge (6). Refer to “Engine Oil Level - Check“ for more information.

17. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i01633611

Engine Valve Lash - Check

SMCS Code: 1105-535

This maintenance is recommended by Caterpillar as part of a lubrication and preventive maintenance schedule in order to help provide maximum engine life.

NOTICE
Only qualified service personnel should perform this maintenance. Refer to the Service Manual or your Caterpillar dealer for the complete valve lash adjustment procedure.

Operation of Caterpillar engines with improper valve adjustments can reduce engine efficiency. This reduced efficiency could result in excessive fuel usage and/or shortened engine component life.

i05400990

Engine Water Pump - Inspect

SMCS Code: 1361-040

A water pump that has failed might cause severe engine overheating. Severe engine overheating could result in the following problems:

- Cracks in the cylinder head
- Piston seizure
- Other potential engine damage

Open the engine compartment.

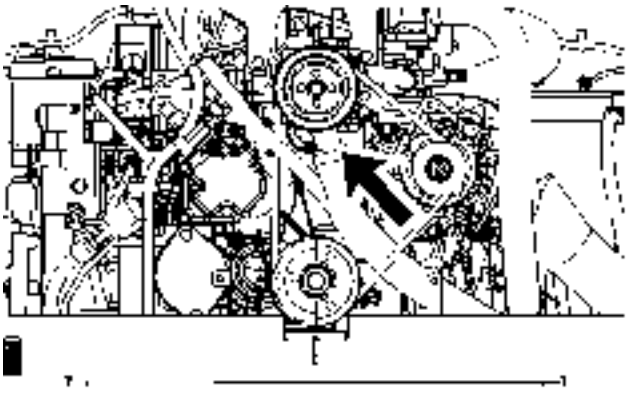


Illustration 119 g03412467

Water pump (1) is located on the engine block at the front of the engine.

Visually inspect the water pump for leaks. If leaks are found, all the seals must be replaced. If there is an excessive leakage of coolant, replace the water pump.

i08192490

Film (Product Identification) - Clean

SMCS Code: 7405-070; 7557-070

Prepare the machine for maintenance. Refer to Operation and Maintenance Manual, “Prepare the Machine for Maintenance”.



Illustration 120 g02174985

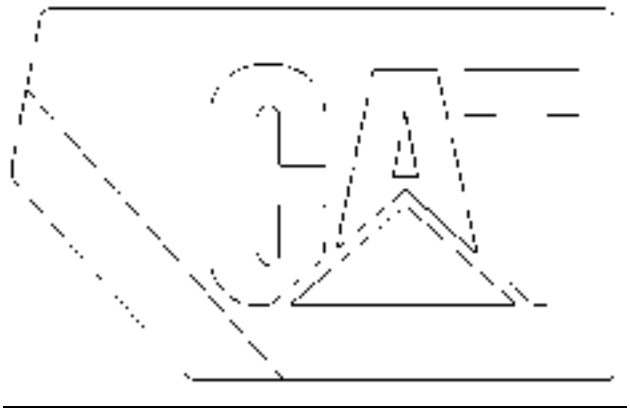


Illustration 121 g02175297



Illustration 122 g06394021

Typical example of the Product Identification Films.

Cleaning of the Films

Make sure that all of the product identification films are legible. Make sure that the recommended procedures are used in order to clean the product identification films. Ensure that all the product identification films are not damaged or missing. Clean the product identification films or replace the films.

Hand Washing

Use a wet solution with no abrasive material that contains no solvents and no alcohol. Use a wet solution with a “pH” value between 3 and 11. Use a soft brush, a rag, or a sponge in order to clean the product identification films. Avoid wearing down the surface of the product identification films with unnecessary scrubbing. Ensure that the surface of the product identification films is flushed with clean water and allow the product identification films to air dry.

Power Washing

Power washing or washing with pressure may be used in order to clean product identification films. However, aggressive washing can damage the product identification films.

Excessive pressure during power washing can damage the product identification films by forcing water underneath the product identification films. Water lessens the adhesion of the product identification film to the product, allowing the product identification film to lift or curl. These problems are magnified by wind. These problems are critical for the perforated film on windows.

To avoid lifting of the edge or other damage to the product identification films, follow these important steps:

- Use a spray nozzle with a wide spray pattern.
- A maximum pressure of 83 bar (1200 psi)
- A maximum water temperature of 50° C (120° F)
- Hold the nozzle perpendicular to the product identification film at a minimum distance of 305 mm (12 inch).
- Do not direct a stream of water at a sharp angle to the edge of the product identification film.

i02625315

Fuel Injector - Test/Change

SMCS Code: 1254-510; 1254-081

WARNING

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

NOTICE
Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over any disconnected fuel system components.

Fuel injectors are subject to tip wear. Tip wear is a result of fuel contamination. Tip wear can cause the following problems:

- Increased fuel consumption
- Black smoke
- Misfire
- Rough running

Fuel injectors should be cleaned, inspected, tested, and replaced, if necessary. Refer to Special Instruction, SEHS7292 for using the 8S-2245 Injector Cleaning Tool Gp. Consult your Caterpillar dealer about cleaning the fuel injectors and testing the fuel injectors.

NOTICE
Never wire brush or scrape a fuel injection nozzle. Wire brushing or scraping a fuel injection nozzle will damage the finely machine orifice. Proper tools for cleaning and testing the fuel injection nozzles can be obtained from Caterpillar dealers.

The following items are symptoms of a malfunction of the fuel injectors:

- Abnormal engine operation
- Smoke emission
- Engine knock

Each fuel injector must be isolated one at a time in order to determine the malfunctioning fuel injector.

1. Start the engine.

2. Loosen each fuel line nut one at a time at the fuel injection pump. A cloth or similar material must be used in order to prevent fuel from spraying on the hot exhaust components. Tighten each nut before loosening the next nut.

3. A faulty fuel injector may be identified when a fuel line nut is loosened and the following conditions are present:

- The exhaust smoke is partially eliminated or the exhaust smoke is completely eliminated.
- Engine performance is not affected.

A fuel injector that is suspected of being faulty should be removed. A new fuel injector should be installed in the cylinder in order to determine if the removed fuel injector is faulty.

Removal and Installation of the Fuel Injection Nozzles

For the removal and the installation of fuel injectors, special tooling is required. Refer to the Service Manual for more information. Consult your Caterpillar dealer for assistance.

i08193365

Fuel System - Prime

SMCS Code: 1250-548

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Dispose of all fluids according to local regulations and mandates.

Use the following procedure to prime the fuel system:

If air enters the fuel system, the air must be purged from the fuel system before the engine can be started. Air can enter the fuel system when the following events occur:

- The fuel tank is empty or the fuel tank has been partially drained.
- The low-pressure fuel lines are disconnected.
- A leak exists in the low-pressure fuel system.
- The fuel filter is replaced.

Mechanical Priming Pump

Primary Filter

Ensure that the air is removed from the primary filter before you prime the secondary fuel filter. Refer to illustration 123 .

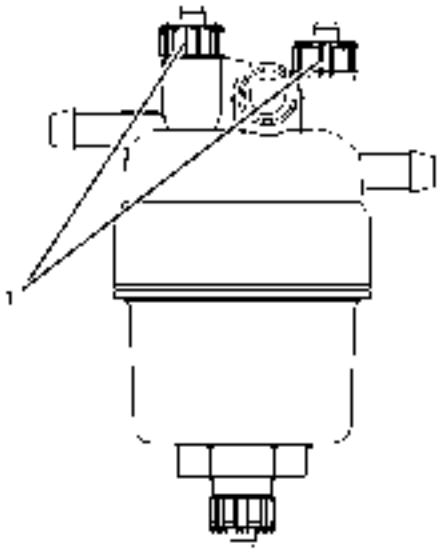


Illustration 123 g01316878

This filter may not be installed on the engine.

(1) Vent screws

Priming the Primary Fuel Filter

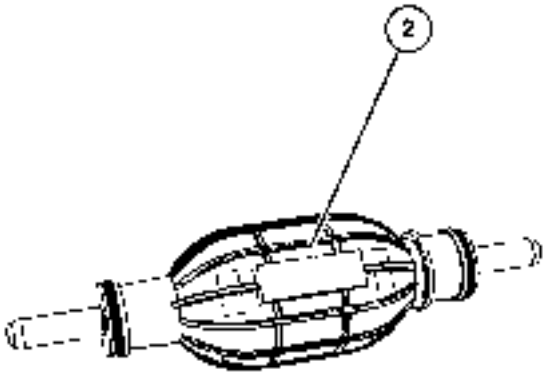


Illustration 124 g06221845

1. If equipped, operate the hand priming pump (2) until fuel free from air comes from the vent screw (1). Then close vent screw (1) by hand.

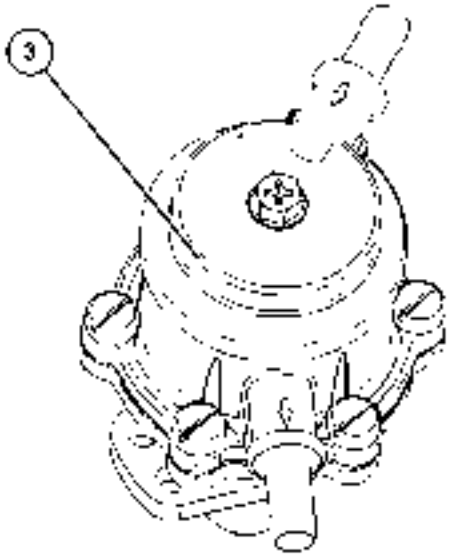


Illustration 125 g06221885

2. If the hand priming pump is not installed, operate the starting motor to crank the engine. Crank the engine to operate the transfer pump (3). Do not operate the transfer pump for more than 15 seconds. After 15 seconds, stop and wait 30 seconds before operating the starting motor.

3. When fuel free from air comes from the vent screw (1) close vent screw (1) by hand.

Secondary Fuel filters

There are three types of fuel filter that may be installed on the engine.

- Element
- Canister
- Spin-on filter with fuel priming pump

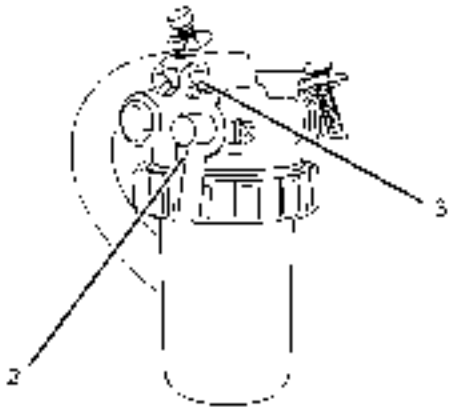


Illustration 126 g01327360

Element

(2) Fuel valve
(3) Vent screw

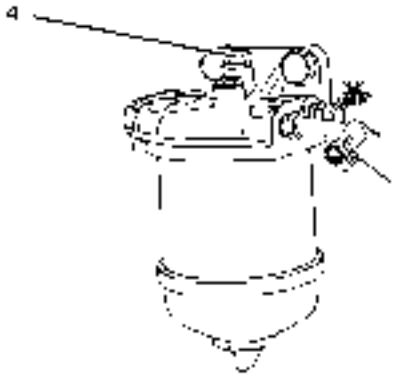


Illustration 127 g01327361

Canister

(4) Vent screw

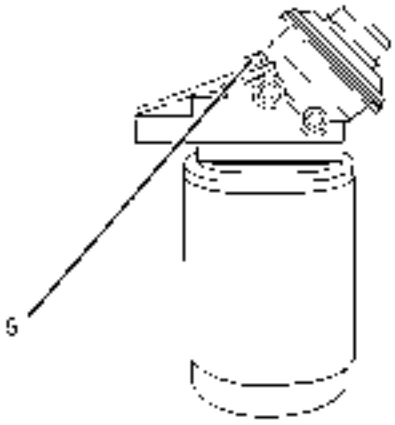


Illustration 128 g01327363

Spin-on filter with fuel priming pump

(5) Vent screw

Vent screw (3) is installed on the filter that has an element. Vent screw (4) is installed on the fuel filter that has a canister. Vent screw (5) is installed on the spin-on filter.

Priming the system

Note: Some fuel system will use gravity to prime the primary fuel filter. When gravity is used to deliver the fuel, ensure that the fuel tank is full and that all stop valves in the fuel line are open.

There are four different types of systems that can be installed on the engine to prime the fuel system. Refer to illustration 129 .

- Hand priming pump
- In-line hand priming pump
- Electrical priming pump
- Transfer pump that is operated by the starting motor

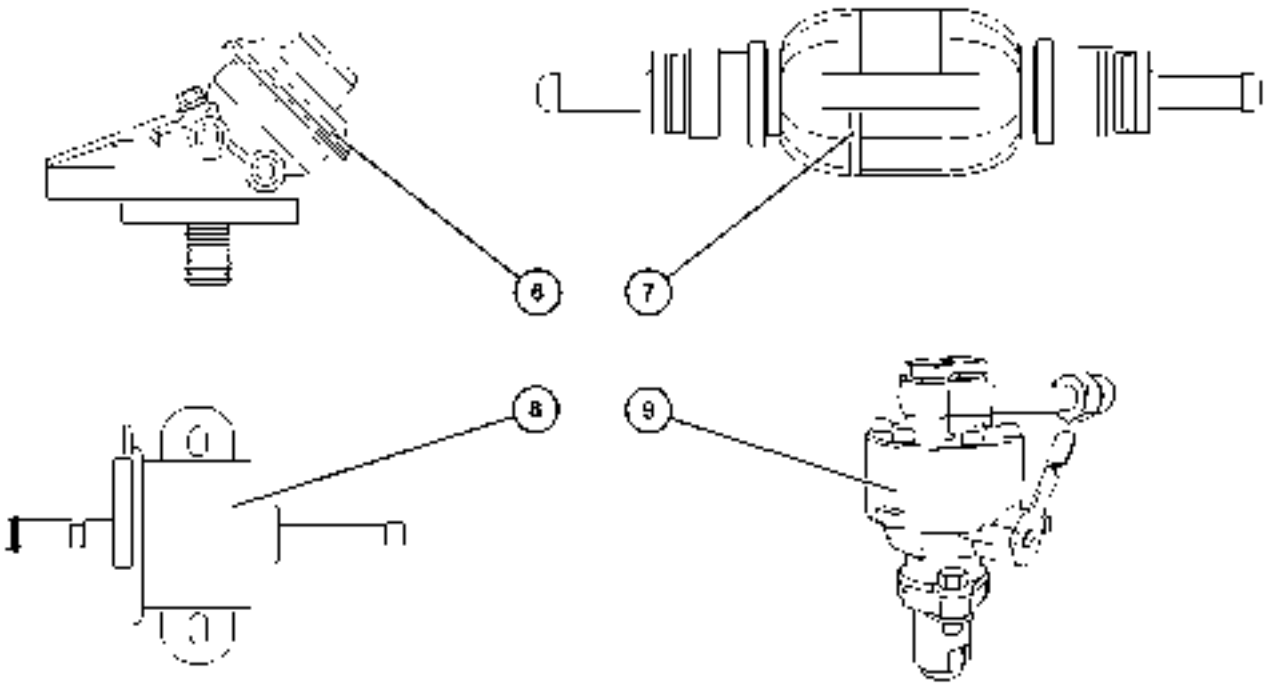


Illustration 129 g06523817

(6) Hand priming pump
(7) In-line priming pump
(8) Electrical priming pump
(9) Fuel transfer pump with hand priming

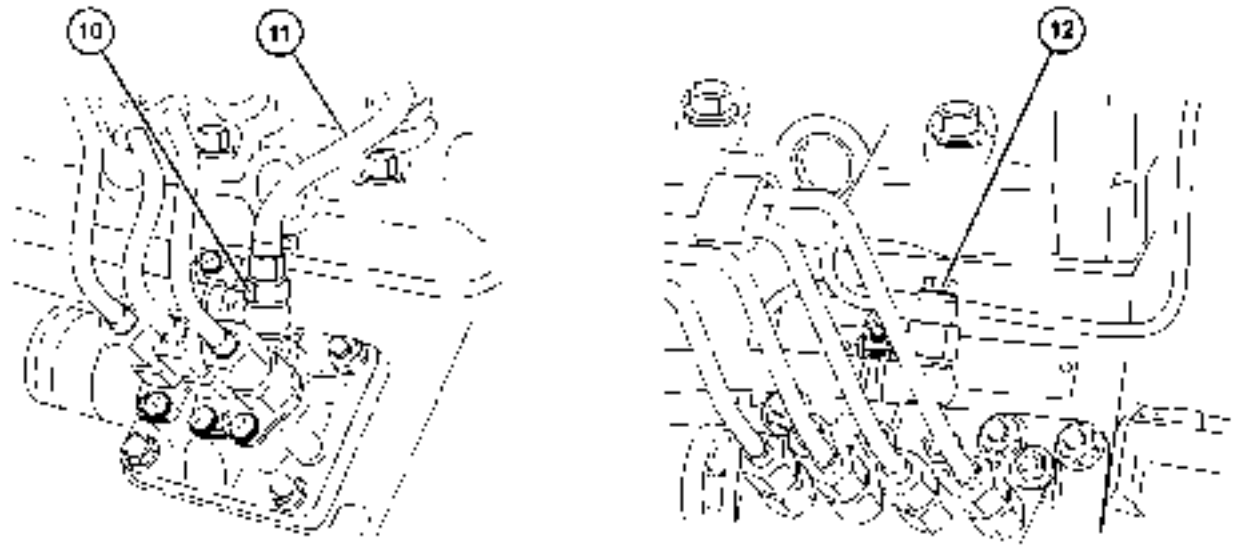


Illustration 130 g06523835

(10) Connector bolt
(11) Fuel return line
(12) Connector bolt

Hand Priming Pump 6

To identify the hand priming pump, refer to illustration 129 .

1. Ensure that fuel valve (2) for the fuel filter that has an element is in the ON position. Refer to illustration 126 .

2. Loosen vent screw (3, 4 or 5) on the fuel filter.

3. Operate hand priming pump (6). When fuel free from air flows from the vent screw tighten the vent screw.

4. Loosen connection (10 or 12) at the fuel injection pump. Refer to illustration 130 .

Note: Fuel return line (11) may need to be removed to prime the fuel system.

5. Operate the hand priming pump. When fuel free from air flows from the connections tighten the connecting bolt.

6. The engine should now be able to start. Operate the starting motor to start the engine.

Note: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

In-line Priming Pump 7

To identify the in-line priming pump, refer to illustration 129 .

1. Ensure that fuel valve (2) for the fuel filter that has an element is in the ON position. Refer to illustration 126 .

2. Loosen vent screw (3, 4 or 5) on the fuel filter.

3. Operate in-line priming pump (7). When fuel free from air flows from the vent screw tighten the vent screw.

4. Loosen connection (10 or 12) at the fuel injection pump. Refer to illustration 130 .

Note: Fuel return line (11) may need to be removed to prime the fuel system.

5. Operate the in-line priming pump. When fuel free from air flows from the connections tighten the connecting bolt.

6. The engine should now be able to start. Operate the starting motor to start the engine.

Note: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

Electrical Priming Pump 8

To identify the electrical priming pump, refer to illustration 129 .

Turn the keyswitch to the ON position for 2 minutes to prime the fuel system. Turn keyswitch to OFF position, then turn on again. The engine is primed and ready to start.

If the engine fails to start, follow Step (1) through Step (6).

1. Ensure that fuel valve (2) for the fuel filter that has an element is in the ON position. Refer to illustration 126 .

2. Loosen vent screw (3, 4 or 5) on the fuel filter.

3. Operate electrical priming pump (8). When fuel free from air flows from the vent screw tighten the vent screw. Switch off the electrical priming pump.

4. Loosen connector bolt (10 or 12) at the fuel injection pump. Refer to illustration 130 .

Note: Fuel return line (11) may need to be removed to prime the fuel system.

5. Operate the electrical priming pump (8). When fuel free from air flows from the connections tighten the connector bolt. If removed, install fuel return line (11).

6. The engine should now be able to start. Operate the starting motor to start the engine.

Note: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

Fuel Transfer Pump 9

To identify the fuel transfer pump, refer to illustration 129 .

Note: If the transfer pump is not equipped with a manual priming lever and the engine is not equipped with any other priming option, you must operate the starting motor to prime the fuel system. Do not operate the starting motor for more than 15 seconds continuously. After 15 seconds, stop and wait for 30 seconds before operating the starting motor again.

1. Ensure that fuel valve (2) for the fuel filter that has an element is in the ON position. Refer to illustration 126 .

2. Loosen vent screw (3, 4 or 5) on the fuel filter.

3. Operate fuel transfer pump (9). When fuel free from air flows from the vent screw tighten the vent screw. Stop the fuel transfer pump.

4. Loosen connection (10 or 12) at the fuel injection pump. Refer to illustration 130 .

Note: Fuel return line (11) may need to be removed to prime the fuel system.

5. Operate the fuel transfer pump. When fuel free from air flows from the connections tighten the connecting bolt. Stop the transfer pump.

6. The engine should now be able to start. Operate the starting motor to start the engine.

Note: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

i08684882

Fuel System Primary Filter - Replace

SMCS Code: 1260-510; 1261-510-PY

WARNING

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire. To help prevent possible injury, turn the start switch off when changing fuel filters or water separator elements. Clean up fuel spills immediately.

WARNING

Personal injury or death can result from a fire.

Fuel leaked or spilled onto hot surfaces or electrical components can cause a fire.

Clean up all leaked or spilled fuel. Do not smoke while working on the fuel system.

Turn the disconnect switch OFF or disconnect the battery when changing fuel filters.

NOTICE
Do not fill fuel filters with fuel before installing them. Contaminated fuel will cause accelerated wear to fuel system parts.

NOTICE
Do not allow dirt to enter the fuel system. Thoroughly clean the area around a fuel system component that will be disconnected. Fit a suitable cover over any disconnected fuel system components.

NOTICE
Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Dispose of all fluids according to local regulations and mandates.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“.

Fuel Filter with Canister

1. Close the fuel supply valve.



Illustration 131 g06734883

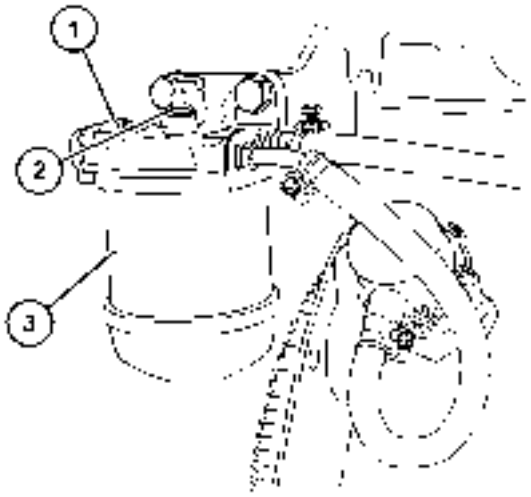


Illustration 132 g06734813

Typical example

(1) Fuel filter assembly
(2) Set screw
(3) Canister

2. Clean the outer surface of fuel filter assembly (1).

3. Remove setscrew (2).

4. Remove canister (3). Ensure that any fluid is drained into a suitable container.

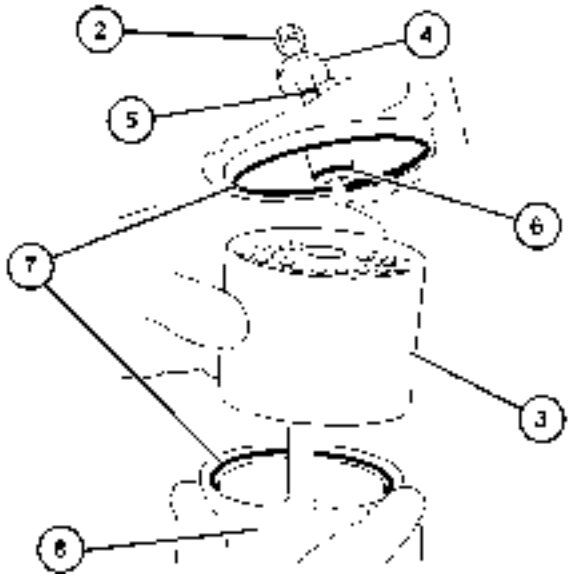


Illustration 133 g06734829

Typical example

(2) Set screw
(3) Canister
(4) Washer
(5) Seal
(6) Seal
(7) Seal
(8) Bowl

5. Assemble the following items: seals (7), seal (6), canister (3), and bowl (8). Place washer (4) and seal (5) on setscrew (2).

6. Fasten the assembly to the fuel filter base with setscrew (2).

The fuel system will need to be primed after the new filter is installed. Refer to “Fuel System - Prime“ for more information.

i05508497

Fuel Tank Strainer - Clean

SMCS Code: 1273-070-STR

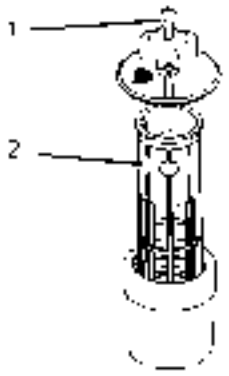


Illustration 134 g02609604

1. Remove fuel tank cap (1).

2. Remove strainer (2) from the filler opening.

3. Wash the strainer in a clean, nonflammable solvent.

4. Install the strainer into the filler opening.

5. Install the fuel tank cap.

i04998176

Fuel Tank Water and Sediment - Drain

SMCS Code: 1273-543-M&S

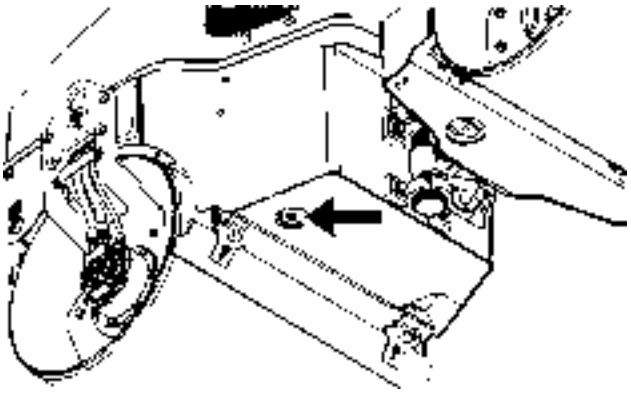


Illustration 135 g03175417

1. The drain plug is located under the left side of the machine.

2. Remove the drain plug. Allow the water and sediment to drain into a suitable container.

3. Install the drain plug.

Note: Dispose of all fluids according to local regulations.

i08470133

Fuses - Replace

SMCS Code: 1417-510

NOTICE
Always replace fuses with the same type and capacity fuse that was removed. Otherwise, electrical damage could result.

NOTICE
If it is necessary to replace fuses frequently, an electrical problem may exist.

Contact your Cat dealer.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“for more information.

Fuse

Fuse – The fuses protect the electrical system from damage that is caused by overloaded circuits. Change the fuse if the element separates. If the element of the new fuse separates, check the circuit. Repair the problem before you operate the machine.

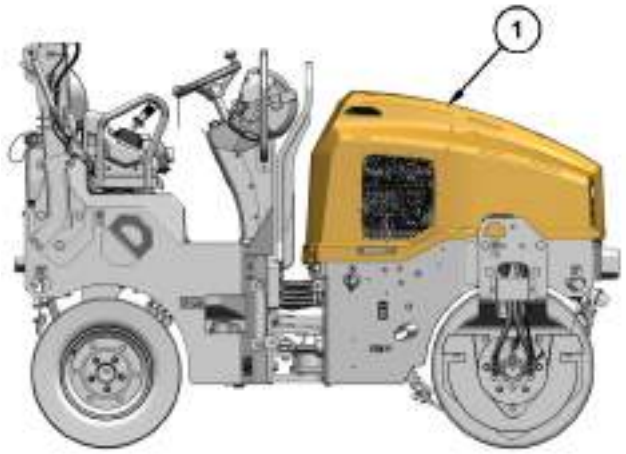


Illustration 136 g06674584

(1) Hood

The fuse box is located below the engine hood (1).

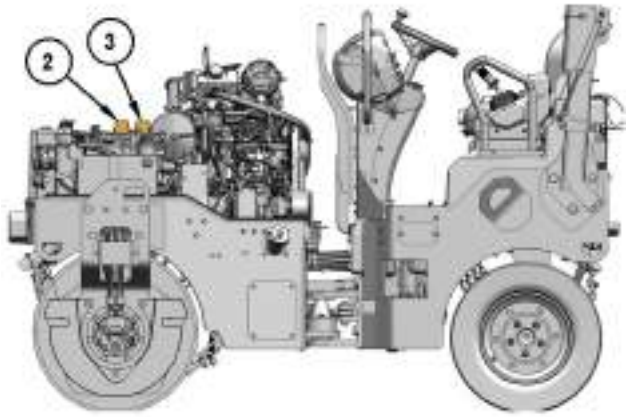


Illustration 137 g06676484

(2) Fuse box 1
(3) Fuse box 2

The fuse boxes are on the top-left side of the machine above the hydraulic tank.

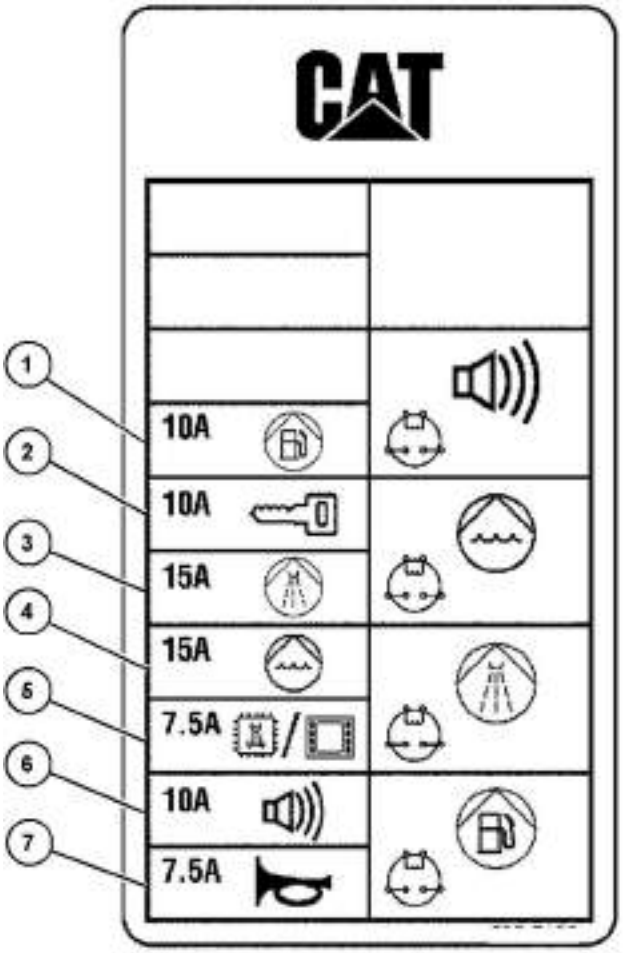


Illustration 138 g06676467

(1) Fuel Gauge – 10 Amp

(2) Key Switch – 10 Amp

(3) Water Spray – 15 Amp

(4) Water Level Gauge – 15 Amp

(5) Product Link / Display – 7.5 Amp

(6) Back Up Alarm – 10 Amp

(7) Horn – 7.5 Amp

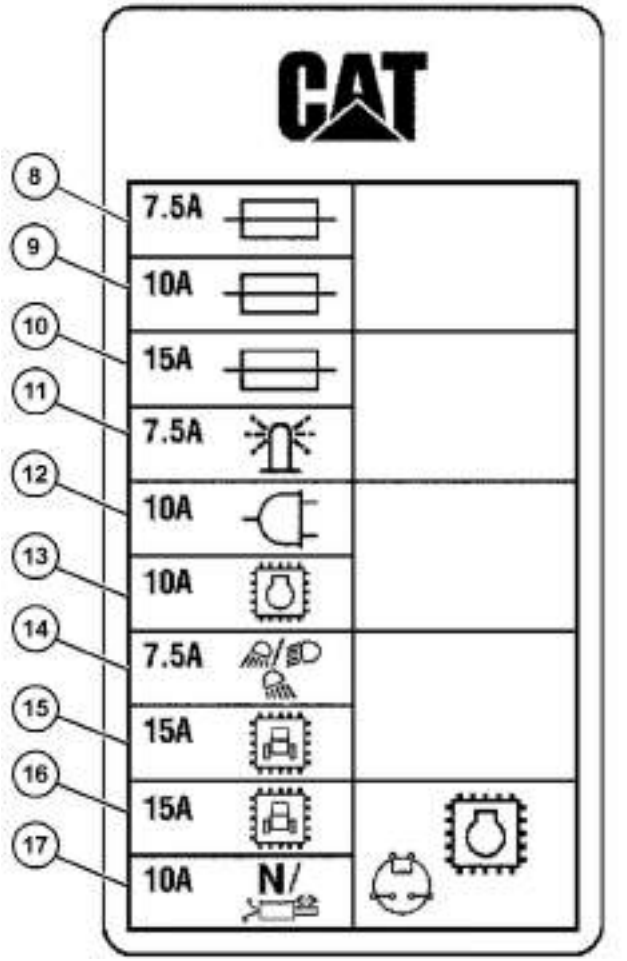


Illustration 139 g06676473

(8) Spare – 7.5 Amp

(9) Spare – 10 Amp

(10) Spare – 15 Amp

(11) Beacon – 7.5 Amp

(12) 12V Power Receptacle – 10 Amp

(13) Engine Electronic Control Module – 10 Amp

(14) Light – 7.5 Amp

(15) Machine Electronic Control Module – 15 Amp

(16) Machine Electronic Control Module – 15 Amp

(17) Neutral Switch – 10 Amp

Relay

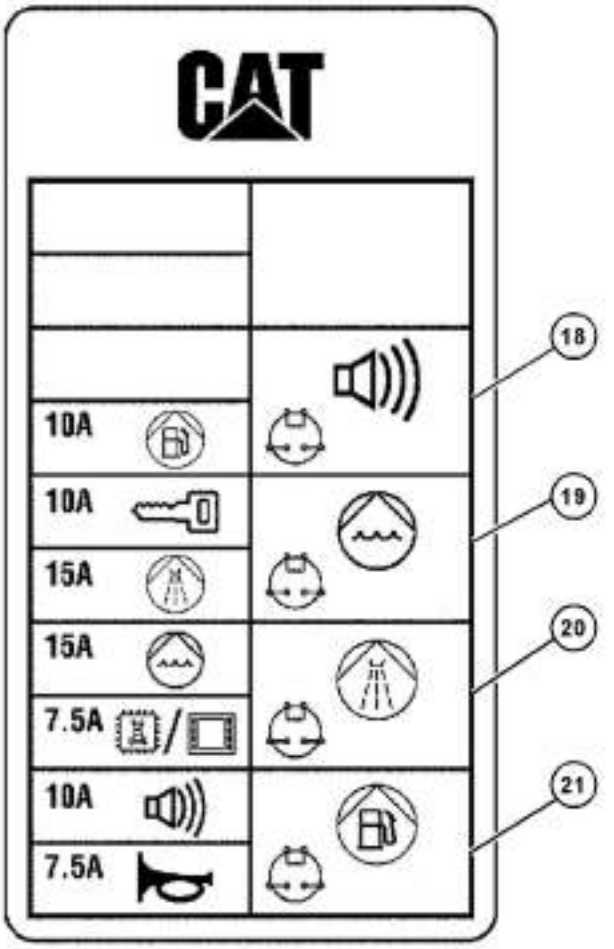


Illustration 140 g06676478

(18) Back Up Alarm – Relay

(19) Water Level Gauge – Relay

(20) Water Spray – Relay

(21) Fuel Gauge – Relay

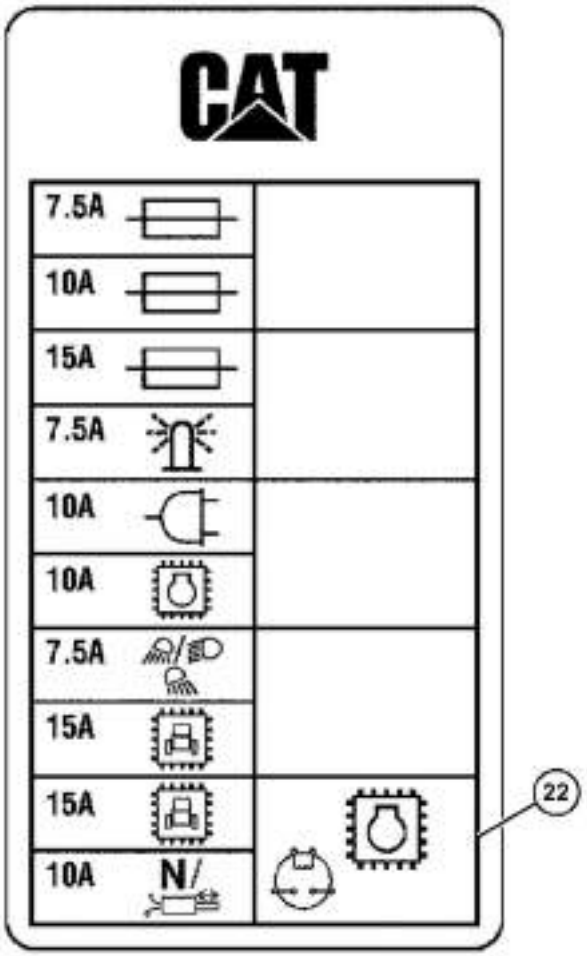


Illustration 141 g06676482

(22) Engine Electronic Control Module – Relay

i08470090

Hydraulic System Oil - Change

SMCS Code: 5050-044; 5095-044

WARNING

Accidental machine starting can cause injury or death to personnel working on the machine.

To avoid accidental machine starting, turn the battery disconnect switch to the OFF position and remove the key. If the machine is not equipped with a battery disconnect switch, disconnect the battery cables from the battery and tape the battery clamps.

Place a do not operate tag at the battery disconnect switch location to inform personnel that the machine is being worked on.

WARNING

At operating temperature, the hydraulic tank is hot and under pressure.

Hot oil and components can cause personal injury. Do not allow hot oil or components to contact skin.

Remove the filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand. Remove the filler cap slowly in order to relieve pressure.

WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the attachments have been lowered, oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

WARNING

Personal injury or death can result from improperly checking for a leak.

Always use a board or cardboard when checking for a leak. Escaping air or fluid under pressure, even a pin-hole size leak, can penetrate body tissue causing serious injury, and possible death.

If fluid is injected into your skin, it must be treated immediately by a doctor familiar with this type of injury.

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact the skin.

NOTICE
Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Cat does not warrant the quality or performance of non-Cat fluids.

NOTICE
Take extreme care to insure the cleanliness of the hydraulic oil. Keep the hydraulic oil clean in order to extend the component life and assure the maximum performance.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

Cat HYDO Advanced 10 Oil Change Interval

The standard Cat ® HYDO Advanced 10 oil change interval is every 3000 service hours or 18 months.

A 6000 service hour or 3 year maintenance interval for hydraulic oil (change) is available. The extended interval requires S·O·S monitoring of the hydraulic oil. The interval for S·O·S monitoring is every 500 hours. The maintenance interval for the hydraulic oil filter is not changed.

Procedure to Change the Hydraulic Oil

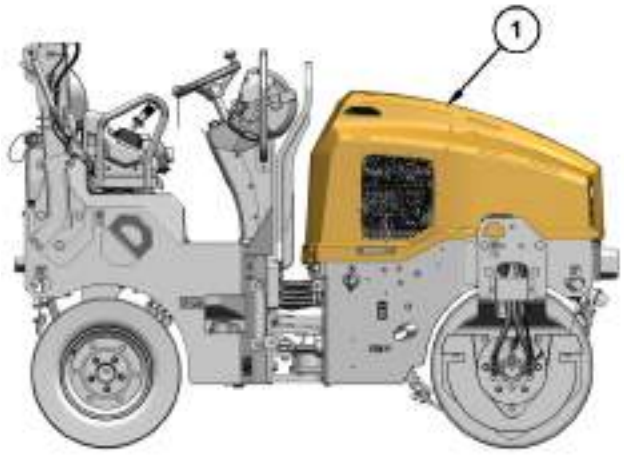


Illustration 142 g06674584
(1) Hood

1. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

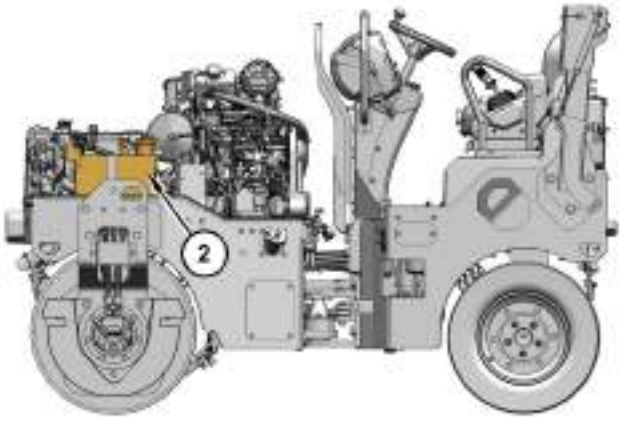


Illustration 143 g06676368

(2) Hydraulic tank

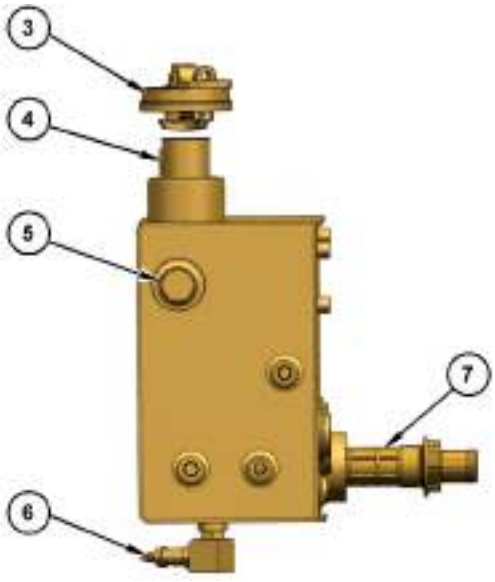


Illustration 144 g06676392

Exploded view of hydraulic tank

(3) Filler cap
(4) Filler screen
(5) Sight gauge
(6) Drain valve
(7) Suction strainer

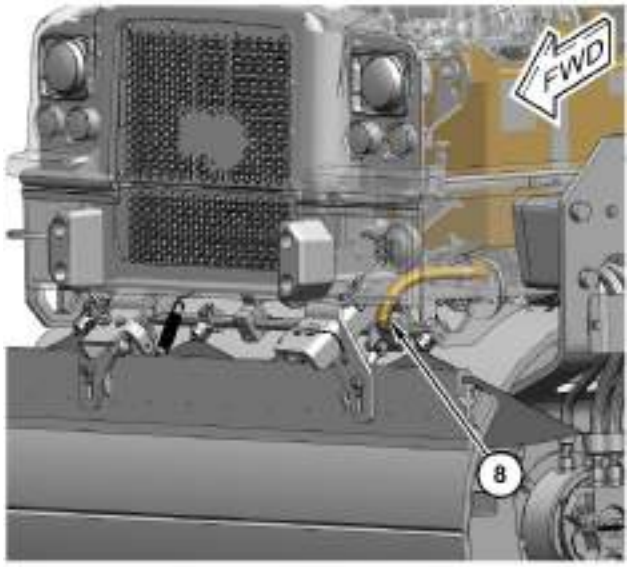


Illustration 145 g06676440

(8) Drain hose

2. Slowly loosen the hydraulic tank filler cap (3) to relieve pressure. Refer to “System Pressure Release“ for more information.

Note: Refer to “General Hazard Information“ for more information that pertains to containing fluid spillage.

3. Open the drain valve (6) to drain the oil from hydraulic tank (2). Place a suitable container under drain hose (8) to collect the oil.

Note: Dispose the drained oil according to local regulations.

4. Close the drain valve (6).

5. Clean hydraulic tank filler cap (3) and the surroundings of filler cap (3).

6. Remove filler cap (3) and filler screen (4). clean filler cap (3) and filler screen (4) by using non-flammable solvent.

7. Remove suction strainer (7) and dispose according to local regulations. Install new suction strainer (7) in hydraulic tank (2).

8. Install filler screen (4) into hydraulic tank (2).

9. Refill hydraulic tank with clean oil. Refer to “Lubricant Viscosities“ and “Capacities (Refill)“ for more information.

10. Check the oil level in sight gauge (5). The oil level should be visible in sight gauge (5). Refer to “Hydraulic System Oil Level - Check“ for more information.

11. Close the filler cap (3). Start the engine and run the engine for few minutes. Refer to “Engine Starting“ for more information.

12. Check the hydraulic tank (2) for any leaks. Maintain the oil level in sight gauge (5). Refer to “Hydraulic System Oil Level - Check“ for more information.

13. Stop the engine. Refer to “Stopping the Engine“ for more information.

14. Close the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

i08470163

Hydraulic System Oil Filter - Replace

SMCS Code: 5068-510

WARNING

Accidental machine starting can cause injury or death to personnel working on the machine.

To avoid accidental machine starting, turn the battery disconnect switch to the OFF position and remove the key. If the machine is not equipped with a battery disconnect switch, disconnect the battery cables from the battery and tape the battery clamps.

Place a do not operate tag at the battery disconnect switch location to inform personnel that the machine is being worked on.

WARNING

At operating temperature, the hydraulic tank is hot and under pressure.

Hot oil and components can cause personal injury. Do not allow hot oil or components to contact skin.

Remove the filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand. Remove the filler cap slowly in order to relieve pressure.

WARNING

Personal injury can result from hydraulic oil pressure and hot oil.

Hydraulic oil pressure can remain in the hydraulic system after the engine has been stopped. Serious injury can be caused if this pressure is not released before any service is done on the hydraulic system.

Make sure all of the attachments have been lowered, oil is cool before removing any components or lines. Remove the oil filler cap only when the engine is stopped, and the filler cap is cool enough to touch with your bare hand.

WARNING

Personal injury or death can result from improperly checking for a leak.

Always use a board or cardboard when checking for a leak. Escaping air or fluid under pressure, even a pin-hole size leak, can penetrate body tissue causing serious injury, and possible death.

If fluid is injected into your skin, it must be treated immediately by a doctor familiar with this type of injury.

WARNING

Hot oil and hot components can cause personal injury. Do not allow hot oil or hot components to contact skin.

NOTICE

Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting, and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids.

Refer to Special Publication, PERJ1017, “Dealer Service Tool Catalog” for tools and supplies suitable to collect and contain fluids on Cat ® products.

Dispose of all fluids according to local regulations and mandates.

NOTICE

Keep all parts clean from contaminants.

Contaminants may cause rapid wear and shortened component life.

NOTICE
Not following the recommendations found in this manual can lead to reduced performance and compartment failure.

Prepare the machine for maintenance. Refer to “Prepare the Machine for Maintenance“ for more information.

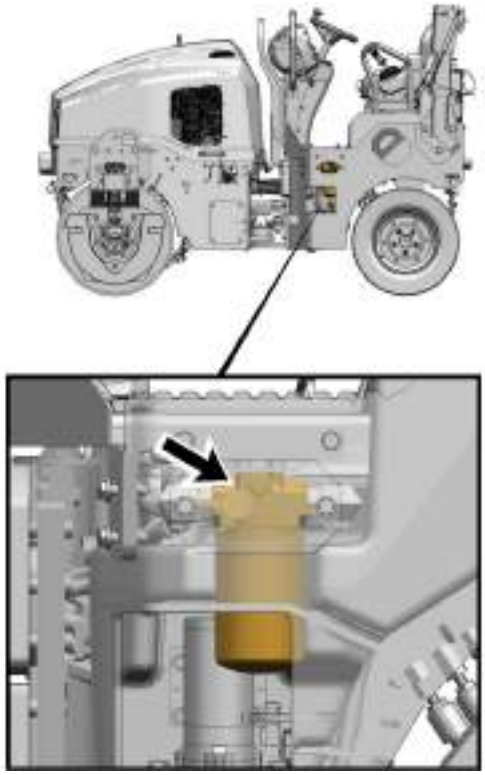


Illustration 146 g06676538

Location of hydraulic system oil filter.

1. The Hydraulic system oil filter is located under the rear frame of the machine. The filter can be accessed through the steps in the rear frame.

2. Clean the surrounding of the filter base. Remove the filter element by using a strap wrench.

Note: Dispose the used filter element according to local regulations.

3. Lubricate the seal of the new filter element with hydraulic oil.

4. Install the new filter element by hand. When the seal contacts the filter base, tighten the filter element by using a strap wrench.

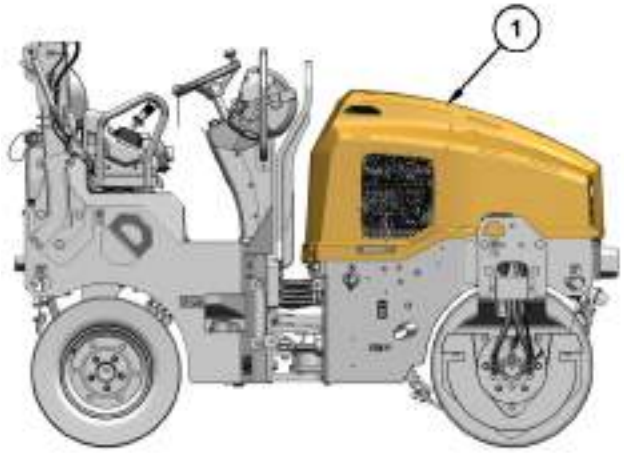


Illustration 147 g06674584

(1) Hood

5. Open the engine compartment hood (1). Refer to “Access Doors and Covers“ for more information.

6. Start the engine and run the engine for few minutes. Refer to “Engine Starting“ for more information.

7. Check for any leaks in the filter and hydraulic lines.